Microscopic Jamming in the Negative Spherical Perceptron

A Ram Calculus Reformulation and Certificate Audit

M413 / Family 222 / L5

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Technical Report Assisted by AI Models

www.integralresearch.net

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Technical Report TR-Ram-OpenAI-722-M413

October 7, 2026

Abstract

This report analyzes the pinned manuscript's route from the margin--1 spherical perceptron with quadratic penalty to gap and force exponents. The main distinction is between a microscopic limit theorem, a global profile-selection argument, and the finite inequalities used inside that argument. The first two remain source claims, not new theorems of this report. We give exact Ram moment and weak-transform reformulations of the profile equation, a rational Bernstein residual certificate, a boundary-aware maximum-principle estimate, and a ground-state test for the shooting problem. Independent finite arithmetic checks cover 9,600 endpoint panels, 24 selected Bernstein conversions, 144 Volterra moment identities, and the exponent conversion. These checks do not replace endpoint tightness, full terminal support, or rigidity of all rescaled critical limits. Private Ram sources were read locally and are cited by their actual passages. No private material was used in public searches or uploaded. An independent bounded audit supplied by the parent additionally verifies four stationary/spectral parameters and proves tighter local error budgets; its scope and attribution are retained separately.

1 Executive Assessment and Claim Discipline

The assignment is mathematically consequential because a stationary scaling ODE and convincing numerical exponents do not, by themselves, construct the microscopic equilibrium law. The pinned manuscript explicitly attempts that construction, including a sharp feasibility threshold and a fixed small non-Gaussian perturbation. Its detailed analytic argument is therefore the object of the study, not merely the polynomial panels in its appendix.

We use four statuses throughout:

Source claim (S).
A theorem or estimate asserted and argued in the pinned manuscript. We trace its prerequisites but do not represent this report as an independent verification of every probabilistic or PDE step.

Conditional transfer (C).
A conclusion follows if the explicitly listed source hypotheses and comparisons hold.

Local proof (L).
A short mathematical result proved in this report on its stated function space or finite interval. No priority claim is made.

Finite check (F).
An executed algebraic test or replay of supplied code. The checked objects, arithmetic, and coverage are specified.

The principal assessment is positive but narrow: Ram's exact-moment, reconstruction-first, weak-adjoint, and stability-ledger machinery fits the *certificate interface* very well. It does not currently replace the microscopic pressure argument, critical martingale compactness, or global classification of entire scaling limits. Calling any finite jet solve a new derivation of the jamming exponents would overstate the evidence.

Two source details are important. The actual pinned file contains 10,356 lines, including the full executable listing. The source's panel residual proof uses an exact integer *power-coefficient ℓ^1 bound*; it is not a Bernstein-basis computation. Section 9 supplies the latter as a separate local reformulation. Also, the margin is $\kappa = -1$, not $+1$.

2 Pinned Object, Model, and Quantifiers

2.1 Provenance and read scope

Reference [S] is the public source at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

Its path is

`preprints/Microscopic-jamming-in-the-negative-spherical-perceptron-September-24-2026/build/main.tex`.

The local file and the raw file obtained from that exact commit have the same SHA-256:

`4d46d60a438d2c7e07c56c5e52ab9c592f58a6af46f8f78365984dc7cdee2be5`.

All substantive source sections, the certificate analysis, bibliography, and complete inline Python listing were read. The adjacent README was also read. No separate external certificate file was present in the local source directory; the appendix was mechanically extracted, without changing its code, to the report directory. Its hash agrees with the hash printed by the source. The directory date is September 24; the text itself records an October 4 complete replay. The commit, not the directory date, fixes the object reviewed.

2.2 Finite system and ordered observables

Fix $\alpha > 0$, $M_N = \lfloor \alpha N \rfloor$, and an $M_N \times N$ random matrix A with independent entries. The state space is $\mathbb{S}_N = \{x \in \mathbb{R}^N : \|x\|^2 = N\}$ with normalized surface measure. Writing $y = x/\sqrt{N}$, define

$$g_\mu = A_\mu y, \quad h_\mu = 1 + g_\mu, \quad V(z) = \tfrac{1}{2}(-1-z)_+^2, \quad H_N = \sum_{\mu=1}^{M_N} V(g_\mu). \tag{1}$$

Thus SAT means that some unit vector satisfies $Ay \geq -\mathbf{1}$. Although the inequalities are linear, the equality $\|y\| = 1$ makes the feasibility problem nonconvex. The Gibbs law at inverse temperature $\beta > 0$ is proportional to $e^{-\beta H_N}$.

For $0 < \delta < u$ and N large enough that $M_N \geq 1$, the gap observable is

$$G_{N,\beta,\alpha}(\delta, u) = \mathbb{E}\left\langle M_N^{-1} \sum_{\mu} \mathbf{1}\{\delta < h_\mu \leq u\} \right\rangle.$$

On $C(x) = \{\mu : h_\mu(x) < 0\}$ define

$$\bar{r} = |C|^{-1} \sum_{\mu \in C} (-h_\mu), \qquad f_\mu = (-h_\mu)/\bar{r}, \quad F_{N,\beta,\alpha}(s) = \mathbb{E}\left\langle |C|^{-1} \sum_{\mu \in C} \mathbf{1}\{f_\mu \leq s\} \right\rangle.$$

The last integrand is zero if C is empty. The ratio is formed for each configuration before the Gibbs and disorder averages; replacing it by a ratio of expectations changes the question.

The order is part of the theorem, not notation that can be rearranged:

$$G_J(u) = \lim_{\delta \downarrow 0} \lim_{\alpha \downarrow \alpha_c} \lim_{\beta \to \infty} \lim_{N \to \infty} G_{N,\beta,\alpha}(\delta, u), \tag{2}$$

$$F_J(s) = \lim_{\alpha \downarrow \alpha_c} \lim_{\beta \to \infty} \lim_{N \to \infty} F_{N,\beta,\alpha}(s). \tag{3}$$

The density approaches from UNSAT. Only after these limits are u and s sent to zero. This is an equilibrium, successive-limit statement, not a finite-N rate, a statement about gradient descent, or a simultaneous scaling theorem in (N, β, α).

2.3 Distributional perturbation and target theorem

The entry law is either standard Gaussian or, for a *fixed* sufficiently small ε, the independent coordinate mixture

$$\nu_\varepsilon = \tfrac{1}{2}\mathcal{N}(0, 1-\varepsilon) + \tfrac{1}{2}\mathcal{N}(0, 1+\varepsilon), \qquad 0 \leq \varepsilon < \varepsilon_0. \tag{4}$$

The second parameter denotes variance. This is not a common random scale per row. Its mean is zero, variance one, and fourth moment $3(1+\varepsilon^2)$. No limit $\varepsilon \to 0$ is taken to obtain the conclusion.

S: main claim. There exist $0 < \alpha_c < \infty$ and $0 < \varepsilon_0 < 1$, common to all such fixed mixtures, such that SAT probability tends to one for every fixed $\alpha < \alpha_c$ and to zero for every fixed $\alpha > \alpha_c$. No assertion at equality is needed. The ordered laws exist, are mixture-independent, and satisfy

$$G_J(u) = u^{1-\gamma+o(1)}, \qquad F_J(s) = s^{1+\theta+o(1)}, \qquad \gamma = \frac{1}{2+\theta}, \tag{5}$$

with

$$.4126930 < \gamma < .4126934, \qquad .4231063 < \theta < .4231088.$$

F_J has total mass and mean one and no atom at zero. The full limiting gap law has a contact atom of mass $1/\alpha_c$. Hence the limiting contact count per degree of freedom is one. This does not say every finite sample has exactly N contacts. The powers in (5) are logarithmic *cumulative* asymptotics; differentiating them into density asymptotics would require additional hypotheses.

3 Current Primary Literature Context

Public-topic searches and primary-page checks were performed on October 7, 2026. No private names, passages, or file contents were included in those queries. This is a focused comparison, not an exhaustive priority search.

Charbonneau, Kurchan, Parisi, Urbani and Zamponi [CKPUZ] develop the full-RSB hard-sphere equations and their jamming predictions. Franz and Parisi [FP] identify nonconvex perceptron criticality and the matching exponent class; they also exhibit a random linear-programming comparison showing that isostaticity alone is insufficient to force the singular distributions. Franz, Parisi, Sevelev, Urbani and Zamponi [FPSUZ] analyze the phase diagram and approach to the SAT–UNSAT transition from both sides. These are antecedents of the physical scaling picture, not evidence that a small panel residual proves a microscopic theorem.

Montanari, Zhong and Zhou [MZZ] establish rigorous feasible and infeasible regions for negative perceptrons and matching leading behavior as the negative margin tends to $-\infty$. That asymptotic regime is distinct from an exact threshold and microscopic laws at the fixed margin -1. Their algorithmic results also concern a different question from the successive equilibrium limits in (2)–(3).

Parisi and Zamponi's June 2026 paper [PZ], published as J. Stat. Mech. 073301, proves a scaling identity within the full-RSB equations. Its abstract explicitly situates the argument in those equations. It is therefore inappropriate to claim the relation in (5) is new here, or to equate that scaling identity with construction and selection of the microscopic limiting profile. The source itself acknowledges this separation. The present report's local reformulations are likewise not a claim of new exponent values or a new universality class. The SciPost landing page was access-blocked during this check; the corresponding primary arXiv record was accessible and used instead.

4 Dependency Analysis: Microscopic Model to Critical Endpoint

The following is a stepwise account of the actual source argument. Every step in this section has status S unless expressly marked otherwise. Listing dependencies exposes what a finite certificate cannot supply.

4.1 Hierarchy, covariance, and the pressure functional

The finite Ruelle probability cascade construction retains independent terminal Gaussian marks for different replicas, including replicas that land on the same leaf. After a change of measure, the genealogy has the required cascade law while the tilted marks factor as stated. This is not an assumption that normalized stable jumps are independent of their total mass. The distinction enters the cavity coefficients.

The row functional $\Phi_u(q;Q)$ has a first-variation formula of the form $-\frac{1}{2}\int(r-q)D_q$, with monotonicity of the martingale-square response D_q. The spherical entropy is written using $\chi_q(t)=\int_t^1 m_q(v)\,\mathrm{d}v$ and a top overlap $q_*<1$:

$$\mathcal{E}(q)=\tfrac{1}{2}\left[\log(1-q_*)+\int_0^{q_*}\chi_q(t)^{-1}\,\mathrm{d}t\right].$$

The inverse spherical response gives the covariance relation. Its concavity, row concavity, and strictness on the relevant support are different statements, each used later. [S, Sections 2–3]

For the row PDE, variance-time coordinates give $F_t=-\frac{1}{2}(F_{zz}+mF_z^2)$. The second variation is organized as a sum of nonpositive squares involving $k=-f_{zz}$ and a modified variation ℓ. This supplies a genuine concavity sign rather than an appeal to the visual shape of a numerical minimizer. Strictness is obtained from hinge curvature on a negative interval. Constants can depend on β. There are two perturbation regimes: a generic terminal perturbation may have a β-dependent admissible radius, whereas a convex perturbation of the loss supported on negative gaps has a fixed small radius. The latter is needed when passing to zero temperature. Truncating the unbounded loss requires a separate cap error and uniform integrability.

For the pressure upper bound, the interpolation sign follows from the row and spherical concavity. For the lower bound, the source uses a block cavity argument, a generic perturbation of scale N^{ρ}, $1/4 < \rho < 1/2$, and synchronization. It first works on appropriate almost-everywhere perturbation-parameter subsequences. Ultrametricity of both coordinates and their sum forces a chain in the joint support; the Ghirlanda–Guerra sampling rules then identify the full common-rank array, not merely a two-replica marginal.

Deleted-row and dimension-increment expansions identify the empirical covariances. The limits in cavity block size and shell thickness have their own order. The argument is not simply superadditivity. The result is the variational pressure $\min_q\{\alpha\Phi(q) + \mathcal{E}(q)\}$, with $q_* < 1$ supplied by a response estimate. Compact row observables are identified by touching, two-sided perturbative bounds; proving a complete formula for every perturbed pressure is not claimed necessary. [S, Section 4]

4.2 Why mixture universality is not moment matching alone

For $\alpha \le B_0$, entry replacement on the diffuse part of the sphere uses $\max_i |y_i| \le N^{-a_0}$ with $1/3 < a_0 < 1/2$. The replacement error before division by N has order N^{2-3a_0}, so this part vanishes after normalization. This estimate does not cover spins concentrated on a few coordinates. The source controls those slices by comparing their entropy loss with the random variance gain on a scale involving $\chi_q(0)$.

One displayed sufficient restriction is

$$\varepsilon_0 < \min\left\{\tfrac{1}{2}, \frac{1}{8(C_0\sqrt{B_0} + B_0)}\right\}, \qquad C_0 = e^2\sqrt{2}(1 + \sqrt{\pi/2}).$$

The crucial point is independence of β, once a bounded density range has been selected. It permits equality of the successive limit functions; it does not justify exchanging their limits. Empirical row convergence, absence of finite-temperature atoms, positivity of negative mass, and residual uniform integrability are needed to pass through the random normalization in F. [S, Section 5]

4.3 Zero energy versus exact satisfiability

The limiting energy $e(\alpha)$ is nondecreasing and Lipschitz, is zero on an initial density interval, and is positive at a sufficiently large finite density. That alone does not identify SAT: an arbitrarily small positive energy can coexist with an infeasible instance.

The source therefore proves uniform geometry for all sufficiently small row subsets, including a lower singular-value bound, and a near-gap count bound. A deterministic active-set repair uses minimum-norm corrections. When new badly affected rows are added, orthogonal increments and an energy budget control the growth. If the corrected norm is too small, an available kernel direction extends the vector while respecting the slabs. If it is too large, radial contraction is compatible with the *negative* margin. Deleting a fixed fraction of rows from a slightly denser zero-energy system leaves enough room for repair. This separate step identifies the sharp threshold with the end of the zero-energy region. [S, Section 6]

4.4 Zero temperature and preservation of small masses

At fixed UNSAT density, the scaled order parameter is $\gamma_\beta = \beta m_q$ and $\bar{\chi} = \beta\chi_q$. Its limit has a bulk density and an endpoint component $\Delta\delta_1$ with $\Delta > 0$. The terminal penalty becomes $-V/(1+\Delta)$. If $H = 1 + Z_1$ is the prelayer field, the physical terminal gap is H when $H \ge 0$ and $H/(1+\Delta)$ when $H < 0$. This layer cannot be ignored when comparing force conventions.

Ground-state Hessian information forces enough active constraints. Uniform control over row subsets combines concentration at order e^{-cN^2} with the at most exponential number of subsets. Low singular values then give estimates relating energy, contact fraction, and Δ. With $L = \bar{\chi}(0)$ one gets $L \asymp e^{-1/2}$ and $\Delta \le Ce^{-1/4}$, hence $\Delta/L \to 0$. This last ratio alone does not exclude an additional concentrated terminal layer. Logarithmic cutoffs, contact-count bounds, and estimates on whole positive-gap windows supply the missing endpoint information. The source treats strictly positive gaps separately from the contact atom. [S, Section 7]

4.5 Critical convex problem and uniqueness of physical limits

On a fixed Wiener space let $R \ge 0$ lie in L^2, and write $r_t = \mathbb{E}[R \mid \mathcal{F}_t] = c + \int_0^t b_v \, \mathrm{d}B_v$. Set $s = 1/\chi$, where $\chi = \Delta + \int_t^1 \gamma(v) \, \mathrm{d}v$, $\gamma \ge 0$ is nondecreasing, and the necessary integrability conditions on s are imposed. The convex functional has the form

$$\begin{aligned} \mathcal{J}_{\alpha,\lambda}(R, s) = \alpha\mathbb{E}[R(1 + B_1) + \tfrac{\lambda}{2}R^2] \\ + \tfrac{\alpha}{2}\left(\frac{c^2}{s(0)} + \int_0^1 \frac{\mathbb{E}b_t^2}{s(t)} \, \mathrm{d}t\right) + \tfrac{1}{2}\int_0^1 s(t) \, \mathrm{d}t. \end{aligned} \tag{6}$$

Quadratic perspectives supply joint convexity, and the susceptibility class is compatible with the corresponding harmonic-mean operation. Positive λ gives strictness in R. The physical normalization is $\lambda = 1/L$, $\chi = \bar{\chi}/L$, and $R = L$ times the residual. The identity $\int s^2 = \alpha\mathbb{E}R^2 = 2e/\lambda^2$ controls the norm.

Weak compactness at criticality is supplemented by strict interior curvature. The equality case is analyzed on the common Wiener space: the Brownian quadratic variation restricts any spatial transformation between two candidates to a translation, and the common normalization removes it. This is a uniqueness argument for the *physical limits constructed by the source*, not a classification of arbitrary formal PDE solutions with unspecified endpoint behavior.

The complementary terminal variables satisfy $H_1 + \Delta R \ge 0$ and $R(H_1 + \Delta R) = 0$. A defect and its primitive obey positivity and complementary support conditions. Their endpoint identity rules out a nonzero critical Δ. A norm squeeze and a critical trial lower bound on $e(\alpha)$ upgrade weak convergence to strong L^2 convergence. Separately, the cutoff argument identifies

$$\mathbb{P}(R > 0) = \mathbb{P}(H_1 = 0) = 1/\alpha_c, \qquad \mathbb{E}R = c > 0. \tag{7}$$

Consequently the limiting force is $R/(\alpha_c c)$ conditional on $R > 0$, and its mean equals $c/[\alpha_c c\mathbb{P}(R > 0)] = 1$. Weak convergence alone would not preserve either this conditional mass or its mean. [S, Section 8]

5 Dependency Analysis: Endpoint Scales to a Selected Profile

5.1 Two scales and terminal support

At criticality normalize $\chi(0) = 1$, put $\tau = 1 - t$, and define

$$w(t) = \int_t^1 \chi(v)^{-2} \, \mathrm{d}v.$$

The source obtains $w(t) \asymp \tau/\chi(t)^2$. Here $\asymp$ means two-sided constants near the endpoint, not an asymptotic ratio of one. For the two endpoint variables $Y \ge 0$ and $R \ge 0$, with $RY = 0$, the positive masses up to the natural scales obey

$$\mathbb{P}(0 < Y \le \sqrt{\tau}) \asymp \chi(t), \qquad \mathbb{P}(0 < R \le \sqrt{w(t)}) \asymp \chi(t). \tag{8}$$

Multiplicative constants do not affect the eventual logarithmic exponents, but excluding the atom at zero is essential.

The proof uses the local field m, $k = -m_h$, $\ell = 1 - k$, and $X = h + m$. Their martingale representations have diffusions proportional to k and ℓ. Heat and Mills-ratio barriers give pointwise bounds. Inverse-flow density estimates first yield rough powers and subpower losses, then uniform scale comparison. Mixed moments such as $\mathbb{E}(rX)$ connect the layer density to the susceptibility and complementarity defect.

In slope coordinates the curvature $\mathcal{V} = -\sqrt{\tau}\, k_h$ is strictly positive and obeys

$$\frac{9}{7}k(1-k)^{3/2} \le \mathcal{V} \le \frac{6}{5}[k(1-k)]^{3/4}, \qquad 0 < k < 1.$$

An inverse-parabolic comparison and a weighted-curvature inequality are used to rule out gaps in the terminal support of the order parameter. On a putative gap, χ would be affine, while the comparison function $\psi = (\alpha \mathbb{E} b^2)^{-1/2}$ has the strict concavity needed for a contradiction at both endpoints. Only after this step may the source differentiate its critical equality to identify

$$A(t) = \frac{\tau\gamma(t)}{\chi(t)} = \frac{\mathbb{E}\mathcal{V}^2}{2\mathbb{E}[k^2(1-k)]}. \tag{9}$$

These are global endpoint arguments, not consequences of the later degree-18 recurrence. [S, Section 9]

5.2 Rescaled evolution and whole-line reference profiles

With $v = \log\tau$, $u = -v$, $z = h/\sqrt{\tau}$, and appropriately rescaled M and density P, the equations are

$$\begin{aligned} M_v &= \tfrac{1}{2}M_{zz} + BM_z + (A - \tfrac{1}{2})M, \\ P_u &= \tfrac{1}{2}P_{zz} - (BP)_z + AP, \qquad B = z/2 + AM. \end{aligned} \tag{10}$$

The time orientations are opposite. Treating both as forward equations in the same time would change the comparison problem. Compactness yields entire limits with $Q = (\log P)_z$ controlled and with integral identities including

$$\int MXP\,\mathrm{d}z = \int k(1-k)P\,\mathrm{d}z = c.$$

The source first restricts A to $[0, .41]$ by a differential inequality and the finite check named `checkA`.

For each $a \in [0, .41]$, the stationary reference profile solves

$$\tfrac{1}{2}M_a'' + (z/2 + aM_a)M_a' + (a - \tfrac{1}{2})M_a = 0, \quad -1 \le M_a' \le 0, \quad M_a'' > 0. \tag{11}$$

It has the prescribed heat/Mills barriers

$$\varphi(z) - z\Phi(-z) \le M_a(z) \le \varphi(z)/\Phi(z),$$

with φ, Φ the standard normal density and CDF. The whole-line existence argument comes from rescalings of a heat-smoothed hinge, and uniqueness uses the damping $1/2 - a > 0$ for bounded differences. The panels approximate this independently defined profile.

Write $B_a = z/2 + aM_a$ and let

$$\mathcal{H}_a = -\tfrac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}z^2} + \tfrac{1}{2}(B_a^2 + B_a') \tag{12}$$

be the Friedrichs realization in $L^2(\mathbb{R})$, initially defined through its quadratic form on $C_c^\infty(\mathbb{R})$. The potential is confining. Its lowest eigenvalue $\lambda(a)$ is simple with a positive eigenfunction. For that eigenfunction ψ_a, set $P_a = e^{\int_0^z B_a}\psi_a$. When $\lambda(a) \ne a$, this reference density satisfies the differentiated logarithmic-density equation but *not* the full stationary equation with eigenvalue a, and it need not satisfy the critical integral identities. That distinction is important in the envelope comparisons.

5.3 Global contraction is an additional certificate target

Using $k = -M'$ and $W = M''$, the inverse reconstruction is

$$M(k) = \int_0^k \frac{j}{W(j)}\,\mathrm{d}j, \qquad X(k) = \int_k^1 \frac{1-j}{W(j)}\,\mathrm{d}j, \qquad z = X - M. \tag{13}$$

The endpoints fix the origin; an arbitrary translation is not available. The source compares all entire limits to upper and lower reference curvatures, including analytic tails and logarithmic-density envelopes. Strict damping in the Q comparison, not only a fitted profile, is used.

The functional

$$\mathcal{N}_\xi = \int_0^1 \frac{P}{W}\left[W^2 + c_+(MX - k(1-k)) - 2\xi k^2(1-k)\right]\mathrm{d}k$$

uses the exact critical identity to have the sign of $A-\xi$. The source's 11 coarse and 13 nonlinear reductions place every admissible entire limit in $[.285, .303]$. Its nonlinear interpolation reference $a = .293650$ is a device for bounding that functional, not an assumed solution of the microscopic scaling limit.

For $d = \sup|A - a_*| < .01$, refined comparisons use

$$L(k) = .26 + 1.30k - 1.86k^2 + 3.25k^3 < 3,$$

giving relative curvature errors $1 \pm dL(k)$, a bound $.65d$ for the profile difference, and $2d$ for logarithmic derivatives. The final functional estimate is $d \le c_{\mathrm{loc}} d$ with $c_{\mathrm{loc}} < .95$. All errors in this step, including omitted tails, must be proportional to d; a fixed small numerical error would prove only a small neighborhood, not $d = 0$. The source explicitly budgets $2 \cdot 10^{-6}$ after division by d. Compactness and uniqueness of the entire limit then yield $A(t) \to a_*$ and

$$\frac{\log \chi(t)}{\log(1-t)} \longrightarrow a_*. \tag{14}$$

This is the key global bridge that a panel-only audit must not silently supply. [S, Section 10]

5.4 Conditional exponent extraction, with proof

Proposition 5.1 (C: common-mass, two-scale conversion)**.** *Assume* (8), (14), $w \asymp \tau/\chi^2$, *and continuity and strict monotonicity of the scale* $\sqrt{w(t)}$ *near the endpoint. If* $0 < a_* < 1/2$, *the cumulative exponents are*

$$1 - \gamma = 2a_*, \qquad 1 + \theta = \frac{2a_*}{1-2a_*}, \qquad \gamma = \frac{1}{2+\theta}. \tag{15}$$

Proof. The first scale is $u = \sqrt{\tau}$, so $\log\chi/\log u \to 2a_*$. Also $\log w/\log\tau \to 1 - 2a_*$; hence for $s = \sqrt{w}$, $\log\chi/\log s \to 2a_*/(1-2a_*)$. Constants in the mass comparisons, conditioning probability, and force normalization disappear upon division by $\log u$ or $\log s$. Continuity and monotonicity of the scales ensure these conclusions hold for every sufficiently small argument, not just a selected sequence. Finally $2 + \theta = 1 + 2a_*/(1-2a_*) = 1/(1-2a_*)$. □

The proof is local algebra once the endpoint statements are granted. The work of constructing and selecting those endpoints remains S.

6 Direct Private Ram Correspondences

The following files were consulted as mathematical sources, not routing maps. They remained local and unchanged. Full absolute locations and hashes are in the accompanying manifest; the descriptions here identify actual passages and their restricted use.

Private source	Passage read	Exact role and limitation
[R208]	TR208, lines 101–595: product-rule jets, monomial shifts, weak RMT adjoints, rank versus PDE stability; lines 752–909: error budgets and adversarial checks.	Supplies the explicit adjoint identity in Section 8. Its finite calibration rank is not a whole-line ODE inverse bound.
[RJ]	`sec-03-jets.tex`, initial exact-moment theorem; nonlinear map and closure, lines 1211–1386; reconstruction consistency and ledger, lines 1387–1555.	Supplies displacement $\alpha = s - t$, exact moments, and the distinction between polynomial forward exactness and closure. One common reconstruction enforces genuine jets.
[RB]	Volume 16, `ch05d-clearance.tex`, full 299-line chapter: Bernstein hulls, dyadic subdivision, and between-node failures.	Supplies the rational power-to-Bernstein transform. A polynomial hull bounds the trial residual, not the true profile without a stability theorem.
[RS]	`stability.tex`, full 295-line file: finite perturbations, boundary realization, residual-ball theorem, and strong monotonicity.	Supplies the exact role of the inverse, its norm and domain. A left inverse for a rectangular residual map does not by itself imply a zero of the original residual.
[RL]	Volume 21, `ch01f-spectral-limits.tex`, full 242-line chapter: spectral-pollution example, graph domains, two-sided inverse tests and dual-output bounds.	Motivates checking a positive whole-line eigenfunction rather than only a small matrix eigenvalue. Its indefinite pollution example does not assert pollution of the lowest Ritz value for this semibounded confining operator.

The correspondence is not "Ram is another name for solving an ODE." It is the explicit localization of this nonlinear operator into exact moments and derivative-consistent panels, followed by a declared analytic inverse estimate and a boundary/tail ledger. The following sections provide the maps and proofs rather than relying on that description.

7 Exact Ram Moment and Jet Reformulation

7.1 A Volterra equation on each physical panel

Fix $a \in [0, .41]$, a nonzero signed panel width h, and $z = z_0 + hs$, $s \in [0, 1]$. Put $y(s) = M(z_0 + hs)$. Twice (11) becomes

$$y'' + h(z_0 + hs + 2ay)y' - h^2(1 - 2a)y = 0. \tag{16}$$

Define the nonlinear source

$$\mathcal{F}[y](s) = -h(z_0 + hs + 2ay(s))y'(s) + h^2(1 - 2a)y(s).$$

For prescribed $y(0) = m_0$ and $y'(0) = v_0$, (16) is equivalent, for $y \in C^2[0,1]$, to

$$y(s) = m_0 + v_0 s + \int_0^s (s-t)\mathcal{F}[y](t)\,\mathrm{d}t. \tag{17}$$

One direction is integration twice; differentiating the right side twice proves the other, with the initial conditions retained. This is a causal Volterra identity, not a global boundary-value solver.

Proposition 7.1 (L: exact Ram moments for the panel Green kernel)**.** *For $f \in C^{n+1}[0,1]$, $0 \le s \le 1$,*

$$\int_0^s (s-t)f(t)\,\mathrm{d}t = \sum_{j=0}^{n} \frac{(-1)^j s^{j+2}}{j!(j+2)} f^{(j)}(s) + \rho_n(s), \tag{18}$$

$$|\rho_n(s)| \le \frac{s^{n+3}}{(n+1)!(n+3)} \left\| f^{(n+1)} \right\|_{L^\infty(0,s)}. \tag{19}$$

Proof. Use the displacement $\alpha = s - t$ specified in [RJ]. The exact kernel moment is $\mu_j(s) = \int_0^s \alpha^{j+1}\,\mathrm{d}\alpha = s^{j+2}/(j+2)$. Taylor's formula for $f(s-\alpha)$ has coefficient $(-1)^j/j!$ and remainder bounded by $\alpha^{n+1}\left\|f^{(n+1)}\right\|/(n+1)!$. Multiplication by α and integration give both formulas. The swept segment is contained in $[0,s]$, so no derivative bound outside the domain has been assumed. □

For a degree-18 trial y, $\mathcal{F}[y]$ has degree at most 35. Consequently (18) with $n = 35$ evaluates its Volterra integral *exactly*. This does not make the trial an exact solution of (17). If $r(s) = y''(s) - \mathcal{F}[y](s)$, then its integral defect is exactly

$$y(s) - m_0 - v_0 s - \int_0^s (s-t)\mathcal{F}[y](t)\,\mathrm{d}t = \int_0^s (s-t)r(t)\,\mathrm{d}t, \tag{20}$$

and is bounded by $s^2\|r\|_\infty/2$. A physical z-residual contributes h^2 times its norm to r. Propagating this local defect through thousands of initial-value panels without a stability argument can amplify it; Section 10 instead uses the global boundary realization.

7.2 Nonlinear jets and honest closure

For a reconstruction $y_h = \sum_{j=0}^{18} c_j s^j$, its components $F_j = y_h^{(j)}$ automatically satisfy $F_j' = F_{j+1}$. Solving independently for 19 numbers at each point does not enforce those identities. The counterexample $F_0 = 0, F_1 = 1$ has no underlying function. This is exactly the consistency issue addressed in [RJ], not an optional numerical detail.

Differentiating the physical residual

$$\mathcal{R}_a[M] = M'' + (z + 2aM)M' - (1-2a)M \tag{21}$$

in a direction v gives

$$D\mathcal{R}_a[M]v = v'' + (z+2aM)v' + \big(2aM' - (1-2a)\big)v. \tag{22}$$

The $2aM'v$ term is essential. In the scaled Volterra equation,

$$D\mathcal{F}[y]v = -h(z_0 + hs + 2ay)v' - 2ahy'v + h^2(1-2a)v.$$

Both formulas follow from the product rule and give an exact Jacobian for polynomial reconstruction. They do not justify dropping a highest derivative of the true nonpolynomial solution. In [RJ]'s

notation, the omitted term in a degree-n closure is $f^{(n+1)}\partial_{F_n}\Phi$. It vanishes for the trial polynomial, not automatically for the solution being approximated.

For a coefficient equation $\widehat{A} = A + E$ with $\|E\| \le \eta < \sigma_{\min}(A)$, the finite error bound contains $(\sigma_{\min}(A)-\eta)^{-1}$. Neither the exact moments nor the use of a polynomial basis produces a lower bound for this singular value. The source's maximum-principle argument supplies a more appropriate continuous stability bound here.

8 Weak Ram Transform and Boundary Realization

Take a smooth test atom on a finite interval $I = [b, c]$, for example

$$\Theta(z) = C\, w((z-d)/W_0)\, \psi_j((z-d)/S_0) e^{\zeta(z-d)},$$

with $W_0, S_0 > 0$ and a smooth window. Define the bilinear Ram weak coefficient $\mathcal{R}_\Theta[f] = \int_I f\Theta \,\mathrm{d}z$, as in [R208]. The normalization C is fixed; no complex conjugation is implicit.

Proposition 8.1 (L: exact weak nonlinear profile identity)**.** *For $M \in C^2(I)$ and $\Theta \in C^2(I)$,*

$$\begin{aligned}\int_I \Theta \mathcal{R}_a[M]\,\mathrm{d}z = &\int_I M\{\Theta'' - z\Theta' - (2-2a)\Theta\}\,\mathrm{d}z - a\int_I M^2\Theta'\,\mathrm{d}z \\ &+ \left[\Theta M' - \Theta' M + z\Theta M + a\Theta M^2\right]_b^c. \end{aligned} \tag{23}$$

Proof. Integrate $\Theta M''$ by parts twice and $\Theta z M'$ once. Use $2MM' = (M^2)'$ for the nonlinear term. Combining the $-\Theta M$ from $(z\Theta)'$ with $-(1-2a)\Theta M$ gives $-(2-2a)\Theta M$. All four boundary terms remain unless their traces vanish. □

For compactly supported atoms in the interior, the last line vanishes. For a common global test across panels, the terms cancel at interfaces if the reconstructed M and M' are continuous. Piecewise C^1 with piecewise C^2 is therefore sufficient for this assembled identity. A derivative jump would generate a distributional contribution from M''. The source's exact C^1 joins remove that contribution; they do not assert C^2 joins.

This gives a useful independent test of signs, Jacobians, and assembly without numerical differentiation of sampled data. It also explains why replacing Θ' by $\zeta\Theta$ is generally wrong: derivatives of the window and of the polynomial atom remain. Those are precisely the symbol-remainder terms retained in [R208]. A finite collection of weak moments, even all zero, need not bound the supremum of the residual. Here the continuous polynomial hull supplies that bound separately.

The corresponding weak linearization follows by replacing M by $M + tv$ in (23) and taking the coefficient of t. It yields the adjoint of (22) with the same trace convention. This is a concrete Ram correspondence, not a generic analogy to weak formulations.

9 Rational Panels and an Exact Bernstein Extension

9.1 What the source actually computes

The source stores $\widetilde{M}(z) = D^{-1}\sum_{n=0}^{18} c_n s^n$ with $D = 10^{36}$, integer c_n, and $z = \epsilon(j+s)/100$, $\epsilon \in \{-1, 1\}$, $j = 0, \ldots, 2399$. The parameter is $a = A/A_D$, with $A_D = 10^6$ for the comparison bank and 10^7 for the two final endpoint tests. The recurrence is integer arithmetic with downward division; it need not produce exact zero residual.

The physical residual (21) has power coefficients $r_n/(A_D D^2)$. The exact test is

$$10^{23} \sum_{n=0}^{36} |r_n| < A_D D^2. \tag{24}$$

Since $0 \le s \le 1$, this proves $\left\| \mathcal{R}_a[\widetilde{M}] \right\|_\infty < 10^{-23}$ on the whole panel, not just at nodes. Degree 36 is harmless zero padding; the nonlinearity has degree at most 35. The joins are

$$c_{0,\mathrm{next}} = \sum_n c_n, \qquad c_{1,\mathrm{next}} = \sum_n n c_n. \tag{25}$$

These are value and s-derivative joins. The physical derivative includes 100ϵ; both arms use the same physical derivative at the origin. The source also tests the small terminal deviations at ± 24 and the trial derivative range. [S, Section 12 and Appendix A]

Proposition 9.1 (L: exact Bernstein residual certificate)**.** *Let $r(s) = \sum_{k=0}^n \alpha_k s^k$ have rational coefficients. Define*

$$\beta_j = \sum_{k=0}^{j} \alpha_k \frac{\binom{j}{k}}{\binom{n}{k}}, \quad j = 0, \dots, n. \tag{26}$$

Then $r(s) = \sum_{j=0}^n \beta_j \binom{n}{j} s^j (1-s)^{n-j}$ and

$$\min_j \beta_j \le r(s) \le \max_j \beta_j, \qquad \|r\|_\infty \le \max_j |\beta_j| \le \sum_k |\alpha_k|. \tag{27}$$

All inequalities can be checked over integers after a common positive denominator is cleared.

Proof. For each k, expand $s^k = s^k(s+1-s)^{n-k}$ and compare coefficients of the Bernstein basis. This gives $s^k = \sum_{j=k}^n \binom{j}{k}\binom{n}{k}^{-1}\binom{n}{j} s^j (1-s)^{n-j}$. Substitute into r to obtain (26). The basis functions are nonnegative and sum to one. Finally $0 \le \binom{j}{k}/\binom{n}{k} \le 1$, so each $|\beta_j|$ is at most $\sum_k |\alpha_k|$. □

Thus the source's ℓ^1 certificate already implies the Bernstein bound; there is no need to invent an additional unproved condition. A direct Bernstein computation can sharpen a pessimistic power bound and support exact dyadic subdivision. Failure of a Bernstein sufficient bound is inconclusive, not evidence that the polynomial violates the desired inequality. This is the same distinction made in [RB]'s clearance theorem.

9.2 A between-node counterexample and its significance

Let $x(s) = \frac{1}{4}s^2(1-s)^2$ and consider the equation $x' = 0$. Its residual $r(s) = \frac{1}{2}s(1-s)(1-2s)$ vanishes at $0, 1/2, 1$, but $x(1/2) = 1/64 \neq 0 = x(0)$. Checking residuals at these three nodes certifies neither a zero residual nor a zero solution. This local polynomial example, adapted from the diagnostic in [RB], is why (24) and (27) are materially stronger than a plotted residual. Conversely, even an identically zero residual for $-u'' + u = 0$ allows $u = Ce^z$ with arbitrary C unless boundary or tail data are fixed. Between-node control and inverse stability solve different problems.

10 From a Residual to a Profile Bound

Lemma 10.1 (L: boundary-aware maximum principle)**.** *Let e be C^1 on $[b, c]$, piecewise C^2 on a finite partition, and solve $e'' + b_1 e' - d_1 e = f$ on each open subinterval. Suppose the coefficients have finite one-sided limits, $d_1 \ge d_0 > 0$, and $|f| \le \eta$. If $|e(b)|, |e(c)| \le \delta$, then*

$$\|e\|_\infty \le \max\{\delta, \eta/d_0\}. \tag{28}$$

Proof. At a positive interior maximum above both proposed bounds, $e' = 0$ and $e'' \le 0$, whereas the equation requires $e'' = d_1 e + f > 0$, a contradiction. At a panel interface, continuity of e' gives $e' = 0$; either one-sided second derivative then yields the same contradiction. Apply the argument to $-e$ for a negative minimum. An arbitrarily small increase of the proposed bound removes equality issues before taking a limit. □

Let M be the whole-line source profile, $\widetilde{M}$ the trial, and $e = M - \widetilde{M}$. Subtracting the two nonlinear equations *exactly* gives

$$e'' + (z + 2aM)e' - \{1 - 2a - 2a\widetilde{M}'\}e = -\mathcal{R}_a[\widetilde{M}]. \tag{29}$$

No quadratic remainder was discarded: it has been incorporated into the drift involving the true M. If $a \in [0, .41]$ and $\widetilde{M}' \le .01$, the damping is at least $1 - 2(.41) - 2(.41)(.01) = .1718 > .17$.

At $z = 24$, the source's analytic tail gives $0 < M(24) \le e^{-(1/2)24^2}/24$; at $z = -24$, putting $p = 1/2 - a$, $0 < M(-24) - 24 \le e^{-p24^2}/(2p\,24)$. Since $p \ge .09$, the worst latter bound is less than $7.2 \cdot 10^{-24}$. Together with the tested 10^{-23} trial endpoint defects, these yield $\delta < 1.72 \cdot 10^{-23}$. Inserting $\eta = 10^{-23}$ and $d_0 = .17$ into (28) gives

$$\left\| M - \widetilde{M} \right\|_{L^\infty[-24,24]} < 10^{-22}. \tag{30}$$

This is a conditional application of the local lemma: it uses the source's existence and analytic tail bounds for M. The finite test does not establish those hypotheses. An integrating-factor argument, using a controlled negative drift area and a short initial interval, then gives the source's derivative error $4 \cdot 10^{-20}$. This derivative bound is separately required in the Riccati coefficients; it does not follow by differentiating (30).

10.1 Independent audit tightening, with its proof mechanism

The separate audit [IA] retains the exact damping .1718 rather than rounding it to .17. Since the boundary error is less than $1.710 \cdot 10^{-23}$ and $10^{-23}/.1718 < 6 \cdot 10^{-23}$, the same maximum principle proves

$$\left\| M - \widetilde{M} \right\|_\infty < 6 \cdot 10^{-23}. \tag{31}$$

For completeness, its derivative improvement can be followed without numerically differentiating a bound. On either arm put $F(x) = \partial_x(M - \widetilde{M})(\epsilon x)$ and $\beta(x) = \epsilon B(\epsilon x)$. The error equation gives

$$F' + 2\beta F = d_1 e - r, \qquad |d_1 e - r| < 1.0082(6 \cdot 10^{-23}) + 10^{-23} < 7.1 \cdot 10^{-23}.$$

The profile barriers give $\beta \ge px - .34$ on the negative arm and $\beta \ge x/2$ on the positive arm, so the total negative area of 2β is less than 1.29. On $[0, .1]$, $\beta \le .38$. Integrating the first-order equation and then integrating F over this short interval yields

$$|F(0)| < \frac{1.2 \cdot 10^{-22} + .005e^{1.29}(7.1 \cdot 10^{-23})}{.1e^{-.076}} < 1.31 \cdot 10^{-21},$$
$$|F(x)| < e^{1.29}[1.31 \cdot 10^{-21} + 24(7.1 \cdot 10^{-23})] < 1.1 \cdot 10^{-20}.$$

The audit proves the exponential comparisons with rational Taylor sums and geometric remainder bounds. These are local improvements conditional on the same analytic profile inputs, not a new or narrower root enclosure.

10.2 What a Ram residual-ball alternative would require

On $I = [-24, 24]$, fix boundary values and use $X = \{v \in C^2(I) : v(\pm 24) = 0\}$ with a declared C^2 norm and $Y = C(I)$ with the supremum norm. A boundary-corrected trial defines a nonlinear residual $F : X \to Y$. The theorem in [RS] would require a bounded isomorphism $B : Y \to X$ and explicit bounds

$$\eta = \|BF(0)\|, \qquad q = \sup_{\|v\|_X \le r} \|I - BDF(v)\|_{X\to X} < 1, \qquad \eta + qr \le r.$$

Then the contraction map $v \mapsto v - BF(v)$ has a unique fixed point in the ball, and injectivity of B makes it a true zero of F. Finite boundary errors have to be lifted into F or enclosed separately. A collocation matrix inverse or a measured Jacobian condition number does not certify this B or its whole-space norm. This report does not claim to have computed such a radii certificate. For the present purpose, (29) is simpler and already aligned with the source.

11 Riccati Tails, Shooting, and Lowest-Mode Selection

11.1 Tail enclosure is a half-line statement

Let $x > 0$, $\epsilon \in \{-1, 1\}$, $\beta_\epsilon(x) = \epsilon B_a(\epsilon x)$, and $S_\epsilon(x) = \partial_x \log P(\epsilon x)$. The spectral equation associated with (12) yields

$$S_\epsilon' = 2\beta_\epsilon S_\epsilon - S_\epsilon^2 + 2(1/2 - ak(\epsilon x) - \lambda). \tag{32}$$

Set $p_+ = 1/2$, $p_- = 1/2 - a$ and

$$c = \lambda/p_\epsilon - 1, \quad d = \frac{c(c-1)}{2p_\epsilon}, \quad e = \frac{(2c-3)d}{2p_\epsilon}, \quad T(x) = c/x + d/x^3 + e/x^5. \tag{33}$$

The source constructs upper and lower barriers $T \pm Kx^{-7}$, $K = (10|e| + 3d^2 + 30)/p_\epsilon$, for $x \ge 12$. Its parameter inequalities imply $|d| < 1.4$, $|e| < 24$ and $K < 3066$. The formal algebraic residual of T has the terms

$$\big[(2c-5)e + d^2\big]x^{-6} + 2de\,x^{-8} + e^2x^{-10},$$

with additional exponentially small profile corrections controlled by the analytic tails. Retaining the identity for $p_\epsilon K$, rather than bounding factors independently, is needed for the asserted signs.

Inward comparison from a finite endpoint tending to infinity constructs the bounded branch and selects it uniquely. Integrating S gives $P \sim Cx^c$; therefore $\psi = e^{-\int^B P}$ decays in the Gaussian confining tail. The other branch is not in L^2. A small residual for a polynomial on $[0, 20]$ is not a substitute for this half-line choice. [S, Section 12, Riccati tail analysis]

11.2 Finite inward propagation and endpoint signs

The source initializes at $x = 20$ using the rational value of T and integrates inward with degree-18 rational panels. The Riccati residual is bounded by 10^{-20}. Incorporating the profile error gives a forcing bound $3 \cdot 10^{-19}$, with an additional $2 \cdot 10^{-9}$ when allowing the full spectral window. If $E = S - \widetilde{S}$, then

$$E' = (2\beta - 2\widetilde{S} - E)E + f.$$

The $|E| < .001$ bootstrap, the cellwise drift lower bounds, and the inward integrating factors control amplification. The source bounds the initial tail error by $2.41 \cdot 10^{-6}$, its global propagated error by

$2.9 \cdot 10^{-5}$, and its central error by $4.5 \cdot 10^{-7}$ over the window. At each shooting value with no spectral-window forcing, the origin error is less than 10^{-10} per arm. The independent audit [IA] sharpens this at each trial's own λ to

$$2.41 \cdot 10^{-6} e^{-20} + 20e^{2}(3 \cdot 10^{-19}) < 5.1 \cdot 10^{-15} \tag{34}$$

per arm. This uses the same source attenuation and forcing allowances; it does not tighten the tested spectral window automatically.

Because the negative arm uses $x = -z$, the matching equation is

$$S_{-}(0) + S_{+}(0) = 0, \tag{35}$$

not the difference. Both positive half-line solutions may first be normalized to the same value at zero. The source tests a negative trial mismatch less than -10^{-9} and a positive one greater than 10^{-9} at the two spectral-window endpoints. The total mismatch error is less than $2 \cdot 10^{-10}$, preserving their signs.

Proposition 11.1 (L: positive matching identifies the lowest mode)**.** *Suppose the source's confining continuous potential in* (12) *has half-line solutions which match to a strictly positive $\psi \in L^2(\mathbb{R})$ in the operator domain and satisfy $\mathcal{H}_a \psi = \lambda \psi$. Then $\lambda = \inf \sigma(\mathcal{H}_a)$.*

Proof. For $f \in C_c^{\infty}(\mathbb{R})$ write $f = \psi g$. Integration by parts, using $-\psi''/2 + U\psi = \lambda\psi$, gives

$$\int \left(\tfrac{1}{2} |f'|^2 + (U - \lambda)|f|^2 \right) \mathrm{d}z = \tfrac{1}{2} \int \psi^2 |g'|^2 \, \mathrm{d}z \geq 0. \tag{36}$$

Density in the Friedrichs form domain extends the lower bound to its closure. Thus $\inf \sigma \geq \lambda$, while the assumed eigenfunction gives the reverse inequality. □

This is a conventional ground-state transform, instantiated here to explain exactly which spectral branch the certificate must select. It is not claimed as a new Ram theorem. The private spectral-limits chapter [RL] warns against treating a small compressed residual as a spectral certificate. Its example uses an indefinite diagonal operator; for this semibounded confining operator, conforming form-domain Ritz approximations of the lowest level instead have the usual upper-bound property. The relevant lesson is domain and tail control, not a blanket assertion of lowest-level spectral pollution.

11.3 A useful conditional monotonicity check

At fixed a, assume differentiability in λ of the selected positive tail branch over a shooting window without a pole. Differentiating (32) gives

$$Z' = (2\beta - 2S)Z - 2, \qquad Z = \partial_{\lambda} S.$$

The admissible tail behavior eliminates the growing homogeneous solution, and hence

$$Z(x) = 2 \int_x^{\infty} \exp\left(- \int_x^t (2\beta(v) - 2S(v)) \, \mathrm{d}v \right) \mathrm{d}t > 0. \tag{37}$$

Indeed differentiating the integral gives the stated ODE; the confining drift makes it convergent and its tail has the required order $1/(px)$. Thus the sum in (35) is strictly increasing in λ on such a pole-free window. This provides a structural sign check and local uniqueness of the shooting zero. It does not prove monotonicity of $\lambda(a) - a$ in a, since changing a changes the entire profile and potential. The source uses continuity plus its entire-limit contraction for the root in a.

12 Arithmetic, Replay, and Independently Executed Checks

12.1 What is inside the public executable certificate

The appendix mixes two kinds of arithmetic. Profile and Riccati panel recurrences, their residual coefficient sums, and many endpoint checks are exact integer or rational computations. The larger comparison functionals use a custom binary64 interval class. The latter pads every operation by $(|x| + 10^{-280})2^{-43}$, encloses elementary functions, and adds a cumulative-sum margin. Decimal mathematical constants must be constructed by the supplied rational constructor when a binary literal would not represent them exactly. The inverse-slope integration grid is dyadic, has 2,048 subdivisions per dyadic band, and uses exact grid denominators; partial cells and analytic endpoints are handled explicitly.

These protections are part of a hand-audited certificate implementation, not a formal proof assistant. Passing the code verifies its assertions on this runtime. A rigorous interpretation additionally depends on the analysis of rounding, interval elementary functions, reduction order, and the analytic inequalities attached to each routine. This distinction is particularly important for `checkA`, the envelope functionals, and `localcheck`, which are not all integer arithmetic.

The source records `checkA()` returning $0.002977541856187091 > 0$ and `localcheck()` returning $0.9196188060510444 < .95$, with local numerator 0.7290900413708653 and denominator 0.7928198472818054. These decimal diagnostic values are *source-recorded*; they are not printed by the unchanged driver. Our replay tests the driver's assertions, not an independent rederivation of those four decimals.

12.2 Extraction and replay provenance

The extracted `certificate_source.py` has SHA-256

`7bd5dbd918d9a48fe4da62044e295239b74f7a1607e2643337ad4d073a85dc7f.`

It agrees exactly with the hash stated in [S]. Assertions are enabled; the driver is run unchanged using the bundled Python runtime. The complete stdout, stderr, return code, elapsed time, and runtime versions are in `_audit/M413-author-certificate.stdout.txt`, `_audit/M413-author-certificate.stderr.txt`, and `_audit/M413-author-replay.json`. The attempted full replay was stopped after approximately ten minutes without a printed checkpoint; it did not complete and is not counted as a successful verification. The nonzero return code in the replay record is the deliberate process termination, not evidence that a mathematical assertion in the certificate failed. No conclusion about the full driver's success or failure follows from this attempt. Completed bounded checks are documented separately.

12.3 Independent rational checks

The separately authored `author_checks.py` extracts the same listing and imports its function definitions without executing the driver again. For the two endpoint parameters it obtains trial coefficients from the source recurrence, but *independently* assembles the physical residual by coefficient convolution. The results are:

	$a = .2936533$	$a = .2936535$
Panels checked	4,800	4,800
Internal joins per arm	2,399	2,399
Maximum power-sum residual (rounded display)	$1.40109 \cdot 10^{-29}$	$1.40988 \cdot 10^{-29}$
Bernstein panels checked	12	12
Maximum Bernstein hull (rounded display)	$1.15628 \cdot 10^{-29}$	$1.19140 \cdot 10^{-29}$

The actual pass/fail comparisons are rational, against 10^{-23}; the displayed decimals are only rounded diagnostics. Both arms also share the exact physical derivative and value at zero. The source's terminal defect assertions execute when generating these saved panels.

The Bernstein checks cover indices $0, 1, 99, 1199, 1999, 2399$ on each arm and each endpoint parameter, exactly 24 panels in total. They are not claimed to be a second all-panel Bernstein implementation. The theorem (27), however, explains why every successful all-panel power-sum check also implies the corresponding hull bound.

The Volterra identity (18) was checked using rational arithmetic for every monomial t^k, $k = 0, \dots, 35$, at $s = 0, 1/7, 1/2, 1$: 144 tests. For such a monomial the exact integral is $s^{k+2}/[(k+1)(k+2)]$. These finite tests supplement, rather than replace, the proof of the identity for general smooth functions.

Finally, exact rational conversion of $a_\ell = 2936533/10^7$, $a_u = 2936535/10^7$ gives

$$1 - 2a_u = .4126930, \qquad 1 - 2a_\ell = .4126934,$$
$$\frac{2a_\ell}{1-2a_\ell} - 1 = \frac{873066}{2063467}, \qquad \frac{2a_u}{1-2a_u} - 1 = \frac{174614}{412693}. \tag{38}$$

The last two values are approximately .4231063544994904 and .4231087030795289, strictly inside the stated force interval. Monotonicity of the rational function for $a < 1/2$ gives the orientation of the bounds. This arithmetic cannot establish that a_* lies between the endpoints; that uses the spectral and selection arguments.

12.4 Independent bounded audit supplied by the parent

Audit [IA] was completed by a separate reviewer, not this report author. The author read its handoff, detailed proofs, results, scalar checks, and source-hash record before integrating the conclusions. Byte-identical copies are retained in the report's `evidence` directory, with source and copy hashes in the freeze manifest. Its execution used CPython 3.10.9 and NumPy 1.23.5, distinct from this author's bundled runtime.

The audit executes `certified` at $a = 0, .41, .2936533, .2936535$ and independently reconstructs the completed polynomial residuals by differentiation and convolution. Its total coverage is 19,200 stationary panels, 48,000 Riccati panels, 19,196 stationary C^1 joins, and 47,976 Riccati C^0 joins. It also performs 32 selected Bernstein conversions, 72 rational reversal evaluations, and 294 tail-parameter checks over all 49 seeds, two arms, and three spectral values. These counts are separate from, and partly overlap, this author's 9,600-panel endpoint checks; they should not be added as disjoint coverage.

The largest reported power-sum residual bounds are approximately $1.68605 \cdot 10^{-29}$ for stationary panels and $1.35938 \cdot 10^{-32}$ for Riccati panels. The decisive comparisons are exact rational/integer tests, not these displayed decimals. Fourteen additional scalar-budget checks use rational exponential enclosures. The $a = 0$ control matches the harmonic-oscillator ground eigenvalue $1/2$ and gives an exactly zero middle trial matching sum, a useful sign check.

The audit's bounded mathematical review found no defect in source lines 8930–9206, conditional on the stated stationary-profile existence, admissibility, and tail inputs. It checked the inward Riccati comparison on the entire spectral window, not merely at three numerical values. It did *not* independently verify the full driver, all profile seeds, the custom interval implementation as a whole, the entire-limit contraction at lines 8504–8810, or the microscopic theorem.

Concrete packaging finding. The standalone path `certificate/original_certificate.py` and the external execution records named in source lines 9466–9497 are absent from the pinned public preprint tree, as checked by [IA] against the public GitHub tree and local files. The documented standalone

command therefore cannot run as shipped. This is a reproducibility packaging defect, not missing algorithm or seed data: the complete appendix listing parses and exactly matches the printed SHA-256. The report's extraction script supplies an explicit recovery procedure without changing the original source. The unavailable external logs cannot be used to independently confirm the source's historical run timing, although their absence does not disprove that run.

12.5 Spectral endpoint output and its logical use

The source's integer spectral seeds have denominator 10^{10} and half-window width $10/10^{10} = 10^{-9}$. They give

$$\begin{aligned} .2936535023 < \lambda(.2936533) < .2936535043, \\ .2936533037 < \lambda(.2936535) < .2936533057. \end{aligned} \tag{39}$$

These two boxes were reproduced by [IA]. The endpoint trial mismatch sums are approximately $-1.01103 \cdot 10^{-8}$ and $9.16031 \cdot 10^{-9}$ at the lower a, and $-9.83361 \cdot 10^{-9}$ and $9.43698 \cdot 10^{-9}$ at the upper a. Their signs exceed the required 10^{-9} margins, including the propagated errors. Thus $\lambda(a) - a$ has opposite signs at the endpoints. Assuming the whole-line realization, tail selection, and finite-to-analytic error transfers above, continuity gives a root $a_* \in (.2936533, .2936535)$. The source's local contraction applies to every such root and proves uniqueness there. This does not assert an unqualified global uniqueness theorem for all formal solutions of the nonlinear profile/eigenvalue system outside the certified regime.

13 A Certificate Ledger and Adversarial Audit

Layer	Evidence or margin	What it does not establish
Panel reconstruction	Integer coefficients; exact C^1 joins; power residual $< 10^{-23}$.	Does not prove existence of the whole-line profile or its boundary asymptotics.
Continuous residual	Exact ℓ^1 coefficient sum, or rational Bernstein hull.	Nodal agreement alone is not equivalent. Hulls of a trial polynomial are not error tubes for an ODE solution.
Profile inverse	Damping $> .17$, tail plus trial boundary defects; maximum principle gives 10^{-22}.	The residual is not an error bound without the boundary realization and damping.
Coefficient derivatives	Source integrating-factor estimate $\lvert M' - \widetilde{M}' \rvert < 4 \cdot 10^{-20}$.	A supremum bound for M cannot simply be differentiated.
Riccati half-lines	$T \pm Kx^{-7}$ for $x \geq 12$, admissible branch, Gaussian-decaying ψ.	An arbitrary inward initial value can select the wrong spectral solution.
Riccati finite interval	Exact residual plus drift propagation and profile error; mismatch error $< 2 \cdot 10^{-10}$.	A raw shooting sign without propagated uncertainty is not a certificate.
Lowest mode	Positive matched L^2 function in Friedrichs domain; identity (36).	A small eigenvalue of a finite matrix does not itself identify this branch.
Entire-limit rigidity	Coarse and nonlinear reductions; final $d \leq c_{\mathrm{loc}} d$, $c_{\mathrm{loc}} < .95$.	A fixed additive tail error would leave only an error neighborhood, not exact stationarity.

Layer	Evidence or margin	What it does not establish
Microscopic selection	Pressure, repair, convex critical uniqueness, contact and force tightness, terminal support.	Neither a replay nor a finite-dimensional inverse proves these probabilistic/PDE inputs.
Exponent conversion	Two endpoint mass scales and rational interval arithmetic.	Does not imply density asymptotics, finite-size rates, or algorithmic universality.

The private error ledger [RJ] separates algebraic residual, field and kernel truncation, nonlinear closure, tails, quadrature, rounding, boundary/interface data, and matrix error before multiplying by a stability constant. Here the exact polynomial computation makes some terms zero for the *trial evaluation*, but not for the entire mathematical problem. In particular: exact Volterra moments eliminate quadrature error for that polynomial integral; exact joins eliminate distributional interface defects; neither eliminates the outer boundary error or the true-profile truncation.

13.1 Highest-value independent review targets

An independent certificate reviewer should first check the exact integer recurrence against (21), including the negative arm's coordinate sign; this report partially duplicates that check. Next come the maximum-principle damping and both boundary tails, then the Riccati barrier signs and inward propagation of initialization errors. These connect the finite arithmetic to actual functions.

For the custom intervals, the most consequential audit targets are the elementary-function enclosures, treatment of mathematical decimal constants, cumulative-sum padding, inverse-coordinate cell bracketing, omitted endpoints, and the error proportional to d in the local contraction. A second interval implementation with explicit directed rounding would be more independent than repeating the same Python program.

The most consequential *analytic* review targets lie outside the executable appendix: the row-concavity strictness under permitted perturbations; the cavity lower bound and empirical covariance identification; uniform geometry in the feasibility repair; the localized-spin estimate uniform in β; endpoint norm and contact-mass tightness; exclusion of terminal support gaps; and compactness plus classification of every entire limit. The report has traced these steps, but does not issue an independent theorem-level correctness verdict on all of them. No demonstrated contradiction in those arguments is alleged.

14 Benefits, Costs, and Concrete Extensions

14.1 Benefits of the exact correspondence

The Ram moment form (18) makes the integration kernel and its remainder explicit. Polynomial nonlinearities can be evaluated exactly without silently Taylor-expanding the nonlinearity itself. Reconstruction-first jets make derivative consistency and signed panel scales auditable. The weak identity (23) supplies an independent sign and boundary check. Bernstein conversion gives a continuous-domain certificate with rational arithmetic and optional exact subdivision. The stability ledger prevents a good residual from being mistaken for a solution theorem. These are practical contributions to reproducibility.

14.2 Costs and nonadvantages

None of these coordinates removes the singular endpoint, the whole-line domain, or the microscopic disorder limit. Large monomial degrees and exact rational denominators can be expensive. Many weak tests may still miss a residual component. A finite singular-value margin depends on chosen coordinates and norms. The source's simple power-sum residual bound is already far below its required threshold; a full Bernstein reimplementation would add little precision there. Greater effort is better spent on independent interval functionals and analytic interfaces. No speedup has been benchmarked against the source, and no claim of dimension-free computation or improved exponent accuracy is made.

14.3 Feasible next investigations

1. Build a second checker which consumes only integer coefficient records and validates all residuals, joins, and rational Bernstein hulls. It should not import the source's interval class or recurrence, so its independence exceeds the current duplicate residual assembly.
2. Reimplement the Riccati tail and mismatch enclosure with directed rounding and a machine-readable error ledger. Verify both signs in (35), the whole-line domain, and positivity before labeling the result a lowest-eigenvalue enclosure.
3. Audit the all-entire-limit contraction with parameterized error coefficients. Every contribution must vanish linearly in d; the report should print these coefficients before taking a supremum.
4. Use (23) as an independent assembly check on all panels, including interface traces. This is an algebraic consistency test, not a replacement for a strong residual bound.
5. Explore interval continuation in a only after certifying the linearized boundary realization. A graph of $\lambda(a) - a$ or a finite difference derivative is not a continuation proof.
6. Treat quantitative finite-size convergence and larger mixture families as new problems. The present ordered limits and fixed-small coordinate mixture theorem do not supply such rates or extensions.

15 Conclusion

The pinned manuscript claims a microscopic equilibrium construction with several genuinely separate analytic stages. Its numerical exponent enclosure is meaningful only when attached to those stages. The Ram correspondences established here give exact local identities and a clear certificate architecture: polynomial reconstruction, continuous residual, boundary-aware inverse estimate, admissible tails, and spectral branch selection. The independent arithmetic checks support the polynomial and conversion portions of that architecture. They are not a proof of the global profile-selection or microscopic universality theorem.

These verified local identities and tighter certificate bounds improve transparency while leaving global selection and universality as explicit dependencies. The result is an auditable Ram reformulation and a partially independently checked finite certificate, with the remaining analytic and implementation obligations stated precisely.

A Read Scope, Reproduction, and Artifact Provenance

The report was authored only in the assigned report directory; its audit files have prefix `_audit/M413-author`, and build products live under `_build/TR-M413-Ram-Jamming`. No catalogue, queue, progress file, shared helper, or original manuscript was edited. Public literature queries concerned jamming,

negative perceptrons, and the identified primary papers only. Private files were opened locally, never sent to web tools.

The public source was read through Sections 1–12 and Appendix A, including the hierarchy and pressure proof, mixture argument, repair, zero-temperature limit, convex critical problem, endpoint scales, profile rigidity, exponent conclusion, analytic certificate, and complete code. The private read scope is recorded in Section 6; the source hashes and exact paths are recorded in `M413-author-manifest.json`.

Reproduction commands, from the report root or with these paths made absolute, are

```
python M413-Perceptron-Jamming/author_checks.py replay
python M413-Perceptron-Jamming/author_checks.py finite
```

The exact interpreter used is the bundled runtime at

```
C:/Users/mural/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/pyt
hon.exe.
```

The PDF export uses the existing, unchanged `_tools/compile_report.py`, with the installed MiKTeX, at most two formatting rounds and three internal TeX passes per round. Every page is rendered for inspection. The parent completed the bounded native requests on this actual report: `open_in_codex` returned `queued`, confirming a queued display request only; `compile_latex_document` returned `compile-failed` with "Unable to find standard directories for platform." No TeX-source diagnostic accompanied that platform error. The installed-MiKTeX PDF export is separate and is not represented as a successful native-editor compilation.

The freeze manifest lists SHA-256 hashes for input files, delivered outputs, authored checks, replay logs, and build/audit products; it also records the read scope, URLs, and changed paths. The manifest does not attempt to hash itself. Its final hash is reported separately to the parent. After the manifest is written, this author performs only read-only verification.

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[IA] Independent bounded certificate audit supplied by the parent, October 7, 2026. Local files M413-independent-audit.txt, M413-independent-proofs.txt, M413-independent-results.json, M413-independent-static.json, M413-independent-sourcehashes.json, and checker/handoff. Frozen copies accompany this report under evidence/.

[R208] Private Ram corpus. TR208, multidimensional variable-coefficient operators. TR208-multidimensional-variable-coefficient-operators.tex. Actual local passages and scope listed in Section 6.

[RJ] Private Ram corpus. RamNonlinearIE-PDE/TR1-lin-nonlin-rev2/sec-03-jets.tex. Exact moments, nonlinear jet map, closure defect, reconstruction and error ledger. Local-only source.

[RB] Private Ram corpus. Ram Calculus Books, Volume 16, chapters/ch05d-clearance.tex. Exact Bernstein and continuous-domain clearance certificates. Local-only source.

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[RL] Private Ram corpus. Ram Calculus Books, Volume 21, chapters/ch01f-spectral-limits.tex. Spectral approximation limits, graph domains, and operator certificates. Local-only source.